AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2024

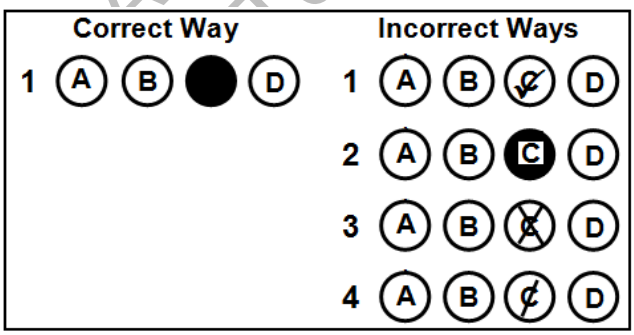
Mathematics Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2 and 3. You may refer to it during the paper, if you wish.
8. You may use a scientific calculator if you wish.

List of Formulae

Note:

- All symbols used in the formulae have their usual meaning.

Complex Numbers

$|z| = \sqrt{a^2 + b^2}$

Matrices and Determinants

$A_{ij} = (-1)^{i+j} M_{ij}$ $\quad AdjA = (A_{ij})^t$ $\quad A^{-1} = \frac{1}{|A|} AdjA$

Sequence & Series and Miscellaneous Series

$a_n = a + (n-1)d$ $\quad A = \frac{a+b}{2}$ $\quad S_n = \frac{n}{2}(2a + (n-1)d)$

$a_n = ar^{n-1}$ $\quad G = \pm\sqrt{ab}$ $\quad H = \frac{2ab}{a+b}$

$S_n = \frac{a(1-r^n)}{1-r}$ if $|r| < 1$ $\quad S_n = \frac{a(r^n - 1)}{r-1}$ if $|r| > 1$ $\quad S_\infty = \frac{a}{1-r}$, where $|r| < 1$

$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ $\quad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ $\quad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$

Permutations, Combinations and Probability

${}^nP_r = \frac{n!}{(n-r)!}$ $\quad {}^nC_r = \frac{n!}{(n-r)!r!}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $\quad P(A/B) = \frac{P(A \cap B)}{P(B)}$ $\quad P(A \cap B) = P(A) \times P(B)$

Binomial Theorem and Mathematical Induction

$(a+x)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}x + \binom{n}{2}a^{n-2}x^2 + \binom{n}{3}a^{n-3}x^3 + \ldots + \binom{n}{n-1}a^1x^{n-1} + x^n$

$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots + \frac{n(n-1)(n-2)\ldots(n-r+1)}{r!}x^r + \ldots$

$T_{r+1} = \binom{n}{r}a^{n-r}x^r$

Quadratic Equation

$x^2 - Sx + P = 0$ $\quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $\quad D = b^2 - 4ac$

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Introduction to Trigonometry and Trigonometric Identities

$l = r\theta$

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \operatorname{cosec}^2\theta$

$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$

$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$

$\tan(\alpha \pm \beta) = \dfrac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$

$\cos\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1+\cos\theta}{2}}$

$\sin\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1-\cos\theta}{2}}$

$\tan\dfrac{\theta}{2} = \pm\sqrt{\dfrac{1-\cos\theta}{1+\cos\theta}}$

$\dfrac{a}{\sin\alpha} = \dfrac{b}{\sin\beta} = \dfrac{c}{\sin\gamma}$

$a^2 = b^2 + c^2 - 2bc\cos\alpha$

$\dfrac{a-b}{a+b} = \dfrac{\tan\frac{\alpha-\beta}{2}}{\tan\frac{\alpha+\beta}{2}}$

$\cos P - \cos Q = -2\sin\dfrac{P+Q}{2}\sin\dfrac{P-Q}{2}$

$\sin P - \sin Q = 2\cos\dfrac{P+Q}{2}\sin\dfrac{P-Q}{2}$

$\cos P + \cos Q = 2\cos\dfrac{P+Q}{2}\cos\dfrac{P-Q}{2}$

$\sin P + \sin Q = 2\sin\dfrac{P+Q}{2}\cos\dfrac{P-Q}{2}$

$\sin\dfrac{\alpha}{2} = \sqrt{\dfrac{(s-b)(s-c)}{bc}}$

$\tan\dfrac{\alpha}{2} = \sqrt{\dfrac{(s-b)(s-c)}{s(s-a)}}$

$\cos\dfrac{\alpha}{2} = \sqrt{\dfrac{s(s-a)}{bc}}$

Application of Trigonometry

$\Delta = \dfrac{1}{2}bc\sin\alpha = \dfrac{1}{2}ac\sin\beta = \dfrac{1}{2}ab\sin\gamma$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$\Delta = \dfrac{c^2\sin\alpha\sin\beta}{2\sin\gamma} = \dfrac{b^2\sin\alpha\sin\gamma}{2\sin\beta} = \dfrac{a^2\sin\beta\sin\gamma}{2\sin\alpha}$

$R = \dfrac{a}{2\sin\alpha} = \dfrac{b}{2\sin\beta} = \dfrac{c}{2\sin\gamma}$

$r_1 = \dfrac{\Delta}{s-a}$, $r_2 = \dfrac{\Delta}{s-b}$ and $r_3 = \dfrac{\Delta}{s-c}$

$r = \dfrac{\Delta}{s}$

$R = \dfrac{abc}{4\Delta}$

Inverse Trigonometric Functions and Trig. Equations

$\sin^{-1}A \pm \sin^{-1}B = \sin^{-1}\left(A\sqrt{1-B^2} \pm B\sqrt{1-A^2}\right)$

$\cos^{-1}A \pm \cos^{-1}B = \cos^{-1}\left(AB \mp \sqrt{(1-A^2)(1-B^2)}\right)$

$\tan^{-1}A \pm \tan^{-1}B = \tan^{-1}\left(\dfrac{A \pm B}{1 \mp AB}\right)$

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1. On simplification of $3i^3 + 4i + 5$, we get

   A. $5 + i$.
   B. $5 - i$.
   C. $2 + 4i$.
   D. $5 - 7i$.

2. If the modulus of a complex number (1, *b*) is 2 then, *b* is equal to

   A. ± 1
   B. ± 3
   C. $\pm \sqrt{5}$
   D. $\pm \sqrt{3}$

3. The multiplicative inverse of $-2i$ is

   A. $\frac{2}{i}$.
   B. $\frac{-2}{i}$.
   C. $\frac{i}{2}$.
   D. $\frac{-i}{2}$.

4. If $Z = 3 + i$, then $\overline{\overline{Z}}$ will be

   A. $3 - i$.
   B. $3 + i$.
   C. $-3 + i$.
   D. $-3 - i$.

5. A square matrix is defined as $M = [m_{ij}]$, where *i* is the number of rows and *j* is the number of columns.

   If matrix *M* is a diagonal matrix, then

   I. $m_{ij} = 0$ for $i \neq j$
   II. $m_{ij} = 0$ for $i = j$
   III. $m_{ij} \neq 0$ for $i = j$

   A. I only.
   B. II only.
   C. I and III.
   D. II and III.

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6. $\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ is equal to

A. $\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$.

B. $\begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$.

C. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.

D. $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.

7. If $4Z$ is a scalar matrix then, Z is a/ an

A. null matrix.
B. identity matrix.
C. diagonal matrix.
D. triangular matrix.

Use the given matrix to answer Q.8 and Q.9.

A matrix A is defined as $A = \begin{bmatrix} 2 & 0 & -4 \\ 4 & -2 & -2 \\ -4 & 2 & 6 \end{bmatrix}$

8. Minor element of a_{23} is

A. -8
B. -2
C. 2
D. 4

9. Cofactor element of A_{32} is

A. 24
B. 12
C. -24
D. -12

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10. Which of the following values of λ will make the matrix $\begin{bmatrix} 3 & 8 & \lambda \\ 1 & 2-\lambda & 1 \\ 0 & 1 & 0 \end{bmatrix}$ singular?

A. 3
B. 0
C. -3
D. -1

11. The determinant of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \end{bmatrix}$ is equal to the product of elements of

A. row 1.
B. column 3.
C. main/principal diagonal.
D. secondary/anti-diagonal.

12. $\begin{bmatrix} 2 & 3 & 1 \\ 1 & -1 & 2 \\ 3 & 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ can be transformed into a system of linear equations as

A. $2x+y+3z=0,\ 3x-y+2z=0,\ x+2y+z=0$
B. $2x+3y+z=0,\ x-y+2z=0,\ 3x+2y+z=0$
C. $2x-y-3z=0,\ 3x+y-2z=0,\ x-2y-z=0$
D. $2x-3y-z=0,\ x+y-2z=0,\ 3x-2y-z=0$

13. If the general term of a sequence is $T_n = \dfrac{n^2(n+1)}{2}$, then the third term is

A. 3
B. 6
C. 12
D. 18

14. The 10th term of the given sequence $\dfrac{a}{2}$, a, $\dfrac{3a}{2}$, $2a$ is

A. $\dfrac{19}{2}a$.
B. $6a$.
C. $5a$.
D. a.

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15. The sum of first five terms of the sequence is 60 and the common difference is equal to the first term.

    The sequence will be

    A. 3, 6, 9…
    B. 4, 8, 12…
    C. 5, 10, 15…
    D. 6, 12, 18…

16. The 3rd geometric mean of 6, GM_1, 48, GM_3, 384 is

    A. $\pm 12\sqrt{2}$
    B. 27
    C. $\pm 96\sqrt{2}$
    D. 216

17. The sum of the infinite sequence 0.6 + 0.06 + 0.006 + … is

    A. $\frac{2}{3}$.
    B. $-\frac{2}{3}$.
    C. $\frac{1}{15}$.
    D. $-\frac{1}{15}$.

18. If the harmonic mean between a and b is 2, then which of the following statements is TRUE?

    A. $a+b=2ab$
    B. $2a+b=ab$
    C. $a+2b=ab$
    D. $2a+2b=2ab$

19. $\sum_{n=1}^{10} n - \sum_{n=1}^{5} n$ is equal to

    A. 0
    B. 2
    C. 40
    D. 55

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20. The sum of the cubes of the first 10 natural numbers is

A. 55
B. 385
C. 2,025
D. 3,025

21. ${}^{8}P_{6}$ is equal to

A. 28
B. 56
C. 20160
D. 67200

22. $\frac{0!(5-1)!}{4}$ is equal to

A. 3!
B. 0!
C. 1
D. 0

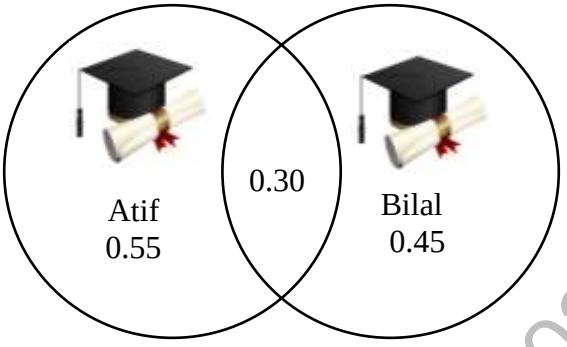
23. ${}^{7}C_{4}+{}^{7}C_{3}$ is equal to

A. ${}^{14}C_{7}$
B. ${}^{8}C_{4}$
C. ${}^{7}C_{7}$
D. ${}^{7}C_{1}$

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24. Atif and Bilal have their final exam in coming week. They may prepare their exam by either studying together or by studying alone.

The Venn diagram shows their probabilities of passing the exam.



The probability that Atif and Bilal study alone and pass the exam is

A. 0
B. 0.3
C. 0.7
D. 1

25. On rolling a pair of balanced/ fair dice, the probability of getting the sum of 5 or 11 will be

A. $\frac{1}{36}$.
B. $\frac{1}{18}$.
C. $\frac{2}{3}$.
D. $\frac{1}{6}$.

26. A bag contains 4 green and 2 red identical balls. If Fareed draws two balls at random from the bag, without replacement, then the probability of drawing two green balls is

A. $\frac{16}{36}$.
B. $\frac{12}{36}$.
C. $\frac{12}{30}$.
D. $\frac{16}{30}$.

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27. The general term of the binomial expansion of $(3x+7y)^{10}$ is

A. $\binom{10}{r}(3x)^{r-10}(7y)^{r}$

B. $\binom{10}{r}(3x)^{r}(7y)^{r-10}$

C. $\binom{10}{r}(3x)^{r}(7y)^{10-r}$

D. $\binom{10}{r}(3x)^{10-r}(7y)^{r}$

28. If the statement $x^{2k} - y^{2k}$ is true for (k + 1) terms, then it must be divisible by

I. $x+y$
II. $x-y$
III. x^2+y^2

A. I only.
B. III only.
C. I and II.
D. II and III.

29. Which of the following conditions is valid for the convergence of $(2+5x)^{-2}$?

A. $|x| < \frac{2}{5}$

B. $|x| > \frac{2}{5}$

C. $|x| < \frac{5}{2}$

D. $|x| > \frac{5}{2}$

30. The coefficient of 3rd term of the expression $(1-2x)^{6}$ will be

A. -60
B. 60
C. -160
D. 160

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31. The solution set of the quadratic equation $2x^2 + \frac{1}{2} = 0$ is

A. $\{-i, i\}$.
B. $\left\{0, \frac{i}{2}\right\}$.
C. $\left\{\frac{i}{4}, -\frac{i}{4}\right\}$.
D. $\left\{\frac{i}{2}, -\frac{i}{2}\right\}$.

32. $(y + 12)(y + 10) = 7$ is a reducible form of

A. $(x+3)(x-4)(x+2)(x-5) = 7$
B. $(x-3)(x+4)(x-2)(x+5) = 7$
C. $(x+3)(x+4)(x+2)(x+5) = 7$
D. $(x+3)(x-4)(x-2)(x+5) = 7$

33. If the roots of the equation $4k - 8y = 20 - y^2$ are equal, then k is equal to

A. 9
B. 6
C. 4
D. 1

34. $\omega^3(\omega^2 - i^2)$ is equal to

A. 1
B. $-\omega$
C. $-\omega^2$
D. ω^2

35. If $y = 2$ and $x^2 - y^2 = 5$, then the solution set of (x, y) is

A. $\{\pm 3, 2\}$.
B. $\{\pm 3, \pm 2\}$.
C. $\{9, \pm 2\}$.
D. $\{\pm 9, 2\}$.

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36. The sum of the roots of the equation $x^2 - (2p+1)x + p = 0$ is equal to four times of the product. The value of p is

A. $-\frac{1}{6}$.

B. $\frac{1}{6}$.

C. $\frac{1}{2}$.

D. $\frac{1}{3}$.

37. 3π radians can be expressed in degrees as

A. 0.16°

B. 9.42°

C. 180°

D. 540°

38. $\sin\theta$ can be represented as

$\frac{\sqrt{\sec^2\theta - 1}}{\sec\theta}$	$\frac{\operatorname{cosec}\theta}{\sqrt{\operatorname{cosec}^2\theta - 1}}$
A	C
$\frac{\sec\theta}{\sqrt{\sec^2\theta - 1}}$	$\frac{\sqrt{\operatorname{cosec}^2\theta - 1}}{\operatorname{cosec}\theta}$
B	D

39. If the distance between (2, 0) and (0, $\sqrt{a}$) is c, then the expression for a will be

A. $c^2 + 4$

B. $-c^2 - 4$

C. $c^2 - 4$

D. $-c^2 + 4$

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40. Which of the following trigonometric ratios has positive sign?

A. $\sin 210^o$
B. $\cos 350^o$
C. $\tan 310^o$
D. $\text{cosec} 240^o$

41. If $3\sin x + 4\cos x = r\sin(x + y)$, then r will be

A. 3
B. 4
C. 5
D. 7

42. $2\sin 4x \cos 3x$ is equal to

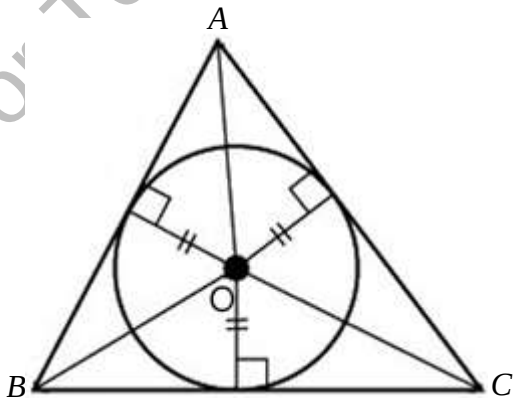
A. $\sin 7x - \sin x$.
B. $\sin 7x + \sin x$.
C. $\sin x - \sin 7x$.
D. $-\sin x - \sin 7x$.

43. The length of hypotenuse of a right angled triangle is three times the length of its base.

If its base is b cm long, then the length of perpendicular will be

A. $\sqrt{8}\,b$.
B. $\sqrt{10}\,b$.
C. $8b$.
D. $10b$.

44. The area of the given equilateral triangle ΔABC is 20.70cm^2 with each side of length 6.9 cm.



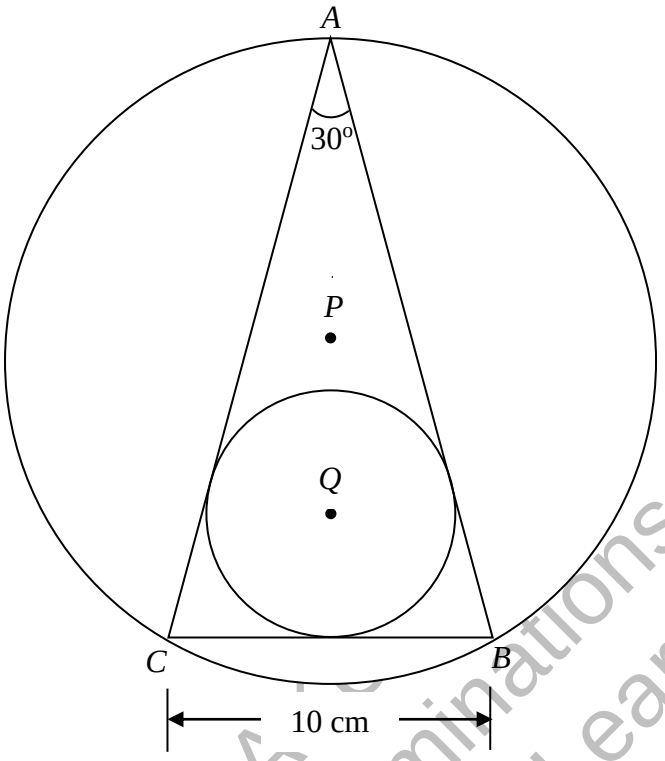
NOT TO SCALE

The in-radius (r) of the triangle ΔABC is

A. 2 cm.
B. 3 cm.
C. 4 cm.
D. 6 cm.

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45. For the given triangle *ABC*, a circum-circle and in-circle are drawn with centres *P* and *Q* respectively. The triangle *ABC* is an isosceles triangle with $BC = 10$ cm and $\angle A = 30^{o}$.



NOT TO SCALE

The area of triangle *ABC* is equal to

A. $50 \times \sin 30^{o}$

B. $100 \times \sin 30^{o}$

C. $100 \times \left(\sin 75^{o}\right)^{2}$

D. $200 \times \left(\sin 75^{o}\right)^{2}$

46. In ΔABC, a = 16 cm, b = 24 cm, c = 20 cm and s = 30. The angle $\cos^{2}\frac{\gamma}{2}$ is equal to

A. $\frac{9}{16}$.

B. $\frac{25}{32}$.

C. $\frac{7}{8}$.

D. $\frac{125}{32}$.

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47. The range of *cosecx* is

A. [-1,1].
B. (-1,1).
C. R - [1,1].
D. R - (1,1).

48. If the period of $f(x) = sin(ax)$ is $\frac{\pi}{2}$, then the value of a is

A. $\frac{1}{4}$
B. $\frac{1}{2}$
C. 2
D. 4

49. The solution set of $3\cot x = \sqrt{3}$ in the interval of $[0,\ 2\pi\ ]$ will be

A. $\left\{\frac{\pi}{3}, \frac{4\pi}{3}\right\}$.
B. $\left\{\frac{\pi}{3}, \frac{7\pi}{6}\right\}$.
C. $\left\{\frac{\pi}{3}+n\pi\right\}\cup\left\{\frac{4\pi}{3}+n\pi\right\}$.
D. $\left\{\frac{\pi}{3}+n\pi\right\}\cup\left\{\frac{7\pi}{6}+n\pi\right\}$.

50. The function $2 - 3\sin(\theta + 40°)$ has the minimum value when θ is equal to

A. 90°
B. 60°
C. 50°
D. 30°

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